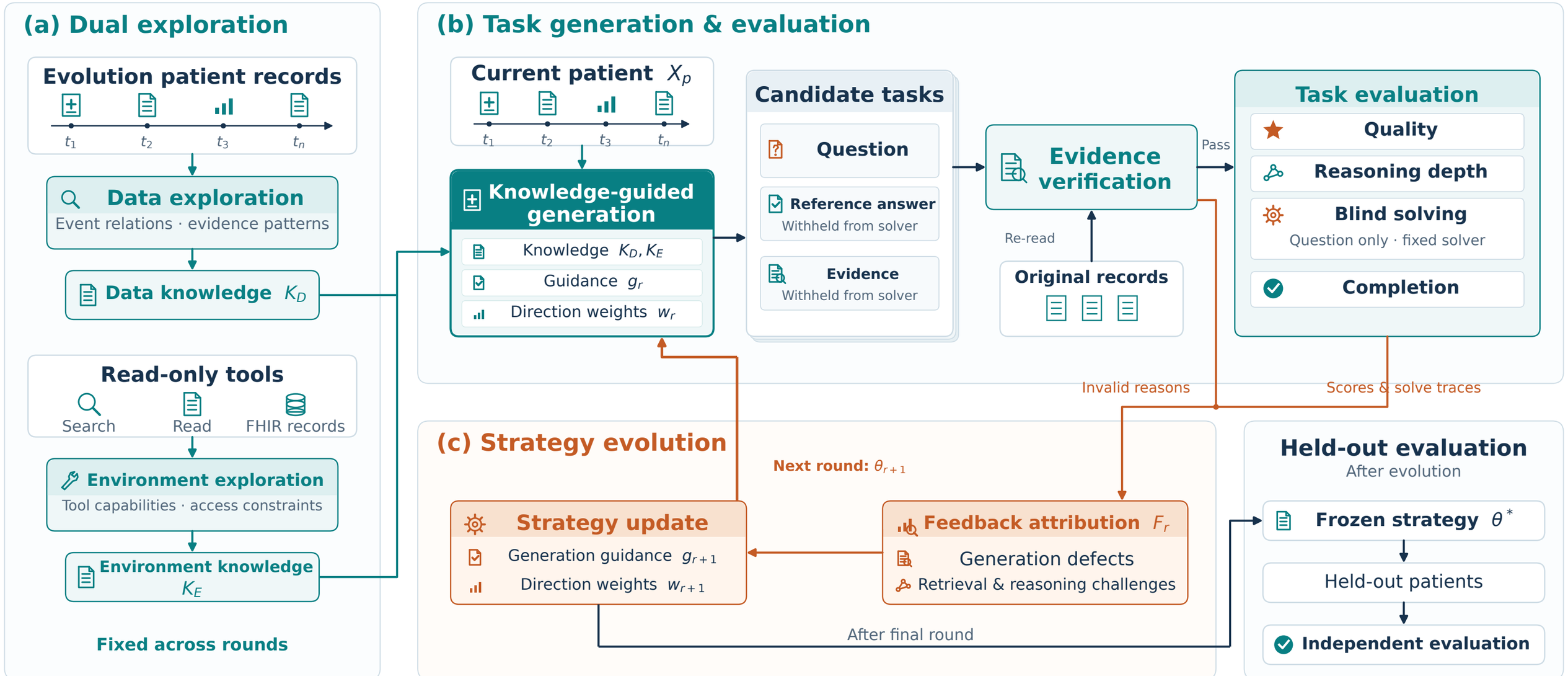

(a) Dual exploration
Evolution patient records
t_1 t_2 t_3 t_n
Data exploration
Event relations · evidence patterns
Data knowledge K_D
Read-only tools
Search
Read
FHIR records
Environment exploration
Tool capabilities · access constraints
Environment knowledge K_E
Fixed across rounds
(b) Task generation & evaluation
Current patient X_p
t_1 t_2 t_3 t_n
Knowledge-guided generation
Knowledge K_D, K_E
Guidance g_r
Direction weights w_r
Candidate tasks
Question
Reference answer
Withheld from solver
Evidence
Withheld from solver
Evidence verification
Pass
Re-read
Original records
Task evaluation
Quality
Reasoning depth
Blind solving
Question only · fixed solver
Completion
Invalid reasons
Scores & solve traces
(c) Strategy evolution
Next round: θ_{r+1}
Strategy update
Generation guidance g_{r+1}
Direction weights w_{r+1}
Feedback attribution F_r
Generation defects
Retrieval & reasoning challenges
After final round
Held-out evaluation
After evolution
Frozen strategy θ^*
Held-out patients
Independent evaluation